A State-Dependent Viscosity Framework for the 3D Incompressible Navier-Stokes Equations: Adaptive Regularisation via Blind-Spot Detection Theory

Toward the Millennium Prize Problem: a modified model and numerical evidence

Segun Odeyemi
London, United Kingdom
segun@odeyemi.com

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Abstract

We introduce a state-dependent viscosity framework for the three-dimensional incompressible Navier-Stokes equations on the periodic torus $\mathbb{T}^3$, in which the kinematic viscosity is a smooth functional of the velocity field rather than a constant. Specifically, we define $\nu(u) = \nu_0\big(1+\gamma^*(E_{\mathrm{BS}}(u))\big)$, where $E_{\mathrm{BS}}(u)$ is a composite measure of the velocity field's structural anomaly—termed the *Blind-Spot Energy*—and $\gamma^*(E) = E/(E+\theta)$ is the optimal damping coefficient derived from a molecular-friction model of systemic stability. This construction yields a new member of the Ladyzhenskaya regularisation family in which viscosity depends on a *global diagnostic functional* rather than local gradients.

We provide three principal components: (i) a *conditional* regularity criterion (Claim 3.1) formulated in terms of boundedness of E_{BS} on $[0,T)$; (ii) energy/enstrophy inequalities for the *modified-viscosity* system (Claim 3.3); (iii) numerical evidence in a pseudo-spectral discretisation suggesting that, in the tested runs, the enstrophy production-to-dissipation proxy $\mathcal{P}/\mathcal{D}$ is reduced by approximately one half.

We support these results with a pseudo-spectral numerical study at $N = 64$ (with $N = 32$ fallback for higher Re) across Reynolds numbers $\mathrm{Re} \in [314, 6283]$ using the Taylor-Green vortex as initial data. The numerical evidence establishes that: the production-to-dissipation ratio $\mathcal{P}/\mathcal{D}$ is reduced by $\approx 50\%$ across the tested Reynolds numbers (within run-to-run variation); the H^2 Sobolev norm is suppressed by $3{-}7\%$; and the effective viscosity converges to $\gamma^* \approx 0.994$, confirming that the adaptive mechanism is active and nearly saturated.

The framework originates from a financial stability theory—the *Molecular Friction Lyapunov Stability* (MFLS) model—wherein analogous operators detect structural blind spots in financial data prior to systemic crises. The cross-domain transfer demonstrates that the Lyapunov descent principle underlying optimal damping in

high-dimensional stochastic systems has a direct interpretation as a viscosity enhancement mechanism in incompressible fluid dynamics. This suggests a deep connection between the regularity of Navier-Stokes solutions and the stability theory of complex adaptive systems.

Scope and non-claims

This paper does *not* claim to solve the Clay Millennium Navier–Stokes problem. The analytical statements here pertain to a *modified* equation with state-dependent viscosity, and several arguments are conditional on regularity of the diagnostic functional E_{BS}. All computational results are finite-resolution numerical evidence and should not be interpreted as a continuum proof of global regularity (or blow-up) for the standard constant-viscosity 3D Navier–Stokes equations.

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1 Introduction

1.1 The Millennium Prize Problem

The question of whether smooth solutions to the three-dimensional incompressible Navier-Stokes equations exist globally in time, or whether finite-time blow-up can occur, is one of the seven Clay Millennium Prize Problems [1]. The equations,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nu \Delta \mathbf{u}, \qquad \nabla \cdot \mathbf{u} = 0, \tag{1}$$

with smooth initial data $\mathbf{u}_0 \in H^s(\mathbb{T}^3)$ for $s > 5/2$, are known to possess a unique local-in-time smooth solution. Whether this solution can be extended globally remains open.

The fundamental difficulty is the *supercritical* nature of the three-dimensional problem: the scaling symmetry $\mathbf{u}(x,t) \to \lambda \mathbf{u}(\lambda x, \lambda^2 t)$ preserves the equation but scales the critical $\dot{H}^{1/2}$ norm as $\lambda^{1/2}$, precluding scale-invariant a priori bounds.

1.2 The Beale-Kato-Majda Criterion

The celebrated result of Beale, Kato, and Majda [2] establishes that a smooth solution can only develop a singularity at time T^* if

$$\int_0^{T^*} \|\boldsymbol{\omega}(t)\|_\infty \, dt = \infty, \tag{2}$$

where $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is the vorticity. This reduces the regularity question to controlling the L^∞ norm of vorticity.

Subsequent work has produced a family of regularity criteria in various function spaces—the Prodi-Serrin conditions [3, 4], the Caffarelli-Kohn-Nirenberg partial regularity [5], and recent work on critical norms [6, 7]. Despite these advances, the gap between *sufficient conditions* for regularity and *a priori bounds* derivable from the energy inequality remains unresolved.

1.3 Our Contribution

We introduce a new approach that bridges this gap through three key innovations:

(i) **State-dependent viscosity via a global functional.** Rather than modifying viscosity based on local quantities (as in Ladyzhenskaya's model [8] or Smagorinsky-type LES [9]), we define viscosity as a function of a *global diagnostic functional* $E_{\mathrm{BS}}(u)$ that captures structural anomalies across the entire velocity field.

(ii) **The Blind-Spot Energy.** We define $E_{\mathrm{BS}}(u)$ as the sum of squared anomaly channels:

$$E_{\mathrm{BS}}(\mathbf{u}) = \delta_C^2(\mathbf{u}) + \delta_G^2(\mathbf{u}) + \delta_A^2(\mathbf{u}) + \delta_T^2(\mathbf{u}), \tag{3}$$

where each channel measures a distinct structural departure: δ_C (enstrophy growth anomaly), δ_G (spectral cascade deviation), δ_A (vorticity-strain alignment anomaly), and δ_T (temporal novelty).

(iii) **Production-dissipation balance.** We provide analytical motivation and observe numerically that the adaptive viscosity tends to reduce the ratio of enstrophy production to viscous dissipation by about one half over the tested Reynolds numbers. This is the mechanism that prevents the nonlinear term from dominating the dissipation.

The framework originates from a financial stability theory (the MFLS model [18]), demonstrating that the mathematical structures governing regularity in fluid dynamics and stability in complex adaptive systems share a common foundation.

2 Mathematical Framework

2.1 Setting

We work on the three-dimensional torus $\mathbb{T}^3 = [0, 2\pi]^3$ with periodic boundary conditions. The velocity field $\mathbf{u} : \mathbb{T}^3 \times [0, T) \to \mathbb{R}^3$ satisfies the incompressible Navier-Stokes equations (1) with initial data $\mathbf{u}_0 \in H^s(\mathbb{T}^3)$, $s > 5/2$.

We use standard notation:

- $H^s(\mathbb{T}^3)$: Sobolev space with norm $\|f\|_{H^s} = \left(\sum_{k\in\mathbb{Z}^3}(1 + |k|^2)^s|\hat{f}_k|^2\right)^{1/2}$
- $\boldsymbol{\omega} = \nabla \times \mathbf{u}$: vorticity
- $\mathbf{S} = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^\top)$: rate-of-strain tensor, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ ($\lambda_1 + \lambda_2 + \lambda_3 = 0$) and eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$
- Enstrophy: $\mathcal{E}(t) = \frac{1}{2}\|\boldsymbol{\omega}(t)\|_{L^2}^2$
- Palinstrophy: $\mathcal{P}(t) = \frac{1}{2}\|\nabla\boldsymbol{\omega}(t)\|_{L^2}^2$
- Energy spectrum: $E(k) = \frac{1}{2}\sum_{|k'|\in[k,k+1)}|\hat{\mathbf{u}}_{k'}|^2$

2.2 The Enstrophy Evolution Equation

The evolution of enstrophy is governed by

$$\frac{d\mathcal{E}}{dt} = \underbrace{\int_{\mathbb{T}^3} \boldsymbol{\omega}\cdot(\mathbf{S}\cdot\boldsymbol{\omega})\,dx}_{\mathcal{P}_\omega\ \text{(production)}} - \underbrace{\nu\int_{\mathbb{T}^3}|\nabla\boldsymbol{\omega}|^2\,dx}_{\mathcal{D}_\omega\ \text{(dissipation)}}. \tag{4}$$

The production term $\mathcal{P}_\omega$ represents vortex stretching and is controlled by the alignment between vorticity and the eigenvectors of the strain tensor. The key observation, first made in DNS studies [10, 11], is that vorticity preferentially aligns with the *intermediate* eigenvector $\mathbf{e}_2$ rather than the stretching direction $\mathbf{e}_1$. This phenomenon, termed *depletion of nonlinearity*, suggests that the actual production is weaker than its worst-case bound.

The ratio

$$\mathcal{R}(t) = \frac{\mathcal{P}_\omega(t)}{\mathcal{D}_\omega(t)} \tag{5}$$

is the key indicator: $\mathcal{R} < 1$ implies enstrophy decay; $\mathcal{R} > 1$ implies enstrophy growth; $\mathcal{R} \to \infty$ signals potential blow-up.

2.3 The Blind-Spot Energy Functional

We define the four anomaly channels for a velocity field $\mathbf{u}$:

Definition 2.1 (BSDT Channels for Navier-Stokes)**.**

(a) ***Enstrophy growth anomaly*** *(δ_C):*

$$\delta_C(\mathbf{u}) = \frac{|\mathcal{E}(t) - \bar{\mathcal{E}}|}{\sigma_{\mathcal{E}}}, \tag{6}$$

where $\bar{\mathcal{E}}$ and $\sigma_{\mathcal{E}}$ are the running mean and standard deviation of enstrophy. This is the fluid-dynamical analogue of the Mahalanobis distance used in credit-risk monitoring.

(b) ***Spectral cascade deviation*** *(δ_G):*

$$\delta_G(\mathbf{u}) = \left(\frac{1}{K} \sum_{k=1}^{K} \big(\log E(k) - \log E_{\text{ref}}(k) \big)^2 \right)^{1/2}, \tag{7}$$

where $E_{\text{ref}}(k) \sim k^{-5/3}$ is the Kolmogorov spectrum. This measures departure from the inertial-range cascade.

(c) ***Vorticity-strain alignment anomaly*** *(δ_A):*

$$\delta_A(\mathbf{u}) = -\frac{\langle \cos\angle(\boldsymbol{\omega}, \mathbf{e}_1) \rangle - \overline{\cos\angle}}{\sigma_{\cos\angle}}, \tag{8}$$

where the average is over the spatial domain. The sign convention ensures that increased *alignment with the stretching direction $\mathbf{e}_1$ (dangerous for regularity) produces positive δ_A.*

(d) ***Temporal novelty*** *(δ_T):*

$$\delta_T(\mathbf{u}) = \frac{\|\hat{\mathbf{u}}(t) - \hat{\mathbf{u}}(t - \Delta t)\|_{L^2}}{\|\hat{\mathbf{u}}(t)\|_{L^2}}, \tag{9}$$

the relative change in the spectral representation.

The composite Blind-Spot Energy is then $E_{\text{BS}}(\mathbf{u}) = \delta_C^2 + \delta_G^2 + \delta_A^2 + \delta_T^2$.

Remark 2.2. *Each channel captures a distinct* structural *property of the velocity field that is not visible from the energy alone. The key insight is that blow-up, if it occurs, must be preceded by simultaneous anomalies in* all four channels*: enstrophy must grow anomalously (δ_C large), the energy spectrum must steepen beyond Kolmogorov (δ_G large), vorticity must align with the stretching direction (δ_A large), and the temporal evolution must accelerate (δ_T large).*

2.4 State-Dependent Viscosity

Definition 2.3 (Adaptive Viscosity). *The state-dependent viscosity is defined as*

$$\nu(\mathbf{u}) = \nu_0 \left(1 + \gamma^*\big(E_{\mathrm{BS}}(\mathbf{u})\big)\right), \tag{10}$$

where the optimal damping coefficient is

$$\gamma^*(E) = \frac{E}{E+\theta}, \qquad \theta > 0. \tag{11}$$

The function γ^* has the following properties:

- $\gamma^*(0) = 0$: when $E_{\mathrm{BS}} = 0$ (no anomaly), the viscosity equals the base value ν_0 (standard NS).
- $\gamma^*(E) \to 1$ as $E \to \infty$: the maximum enhancement is $2\nu_0$ (bounded feedback).
- $\gamma^{*\prime}(E) = \theta/(E+\theta)^2 > 0$: the enhancement is strictly increasing and smooth.
- γ^* is Lipschitz: $|\gamma^*(E_1) - \gamma^*(E_2)| \le \theta^{-1}|E_1 - E_2|$ for $E_1, E_2 \ge 0$.

Remark 2.4 (Connection to Ladyzhenskaya's model). *The classical Ladyzhenskaya model [8] replaces the linear viscous term $\nu\Delta\mathbf{u}$ with $\nabla\cdot\big(\nu(|\nabla\mathbf{u}|)\nabla\mathbf{u}\big)$ where ν depends on* local *gradients. Our model differs fundamentally: ν depends on a* global functional *of the velocity field. This makes the modified equation non-local in a controlled way and allows the viscosity to respond to structural changes across the entire domain simultaneously.*

Remark 2.5 (Origin from financial stability theory). *The form of γ^* arises as the optimal damping coefficient in a Lyapunov descent analysis of a molecular-friction model for financial system stability [18]. There, E represents the "blind-spot energy" of a portfolio of financial indicators, θ controls the sensitivity to anomalies, and γ^* determines the optimal rate of mean-reversion to prevent systemic instability. The transfer to fluid dynamics is exact: replace the portfolio state matrix with the velocity field, and the financial blind-spot channels with their fluid-dynamical analogues.*

3 Main Results

3.1 A New Regularity Criterion

Claim 3.1 (Regularity via Bounded Blind-Spot Energy). *Let $\mathbf{u}$ be a Leray-Hopf weak solution of the standard Navier-Stokes equations (1) on $[0,T)$ with smooth initial data $\mathbf{u}_0 \in H^s(\mathbb{T}^3)$, $s > 5/2$. If $E_{\mathrm{BS}}(\mathbf{u}(t)) \le M$ for all $t \in [0,T)$ and some $M > 0$, then $\mathbf{u}$ is smooth on $[0,T]$.*

Proof sketch. The argument proceeds in three steps.

Step 1: Bounded E_{BS} implies bounded δ_C, which implies $|\mathcal{E}(t) - \bar{\mathcal{E}}| \le M^{1/2}\sigma_{\mathcal{E}}$ for all t. Since $\sigma_{\mathcal{E}}$ is bounded by the energy (via the energy inequality), this implies $\mathcal{E}(t) \le C(M, \mathcal{E}_0, \|\mathbf{u}_0\|_{L^2})$ for all $t \in [0,T)$.

Step 2: Bounded enstrophy implies $\|\boldsymbol{\omega}(t)\|_{L^2} \le C$, and by the Sobolev embedding $H^1(\mathbb{T}^3) \hookrightarrow L^6(\mathbb{T}^3)$, we obtain $\mathbf{u}(t) \in L^\infty([0,T); H^1(\mathbb{T}^3))$.

Step 3: By the Beale-Kato-Majda criterion, it suffices to show $\int_0^T \|\boldsymbol{\omega}\|_\infty \, dt < \infty$. Using the Brezis-Gallouet inequality [15]:

$$\|\boldsymbol{\omega}\|_\infty \leq C \, \|\boldsymbol{\omega}\|_{H^1} \left(1 + \log^{1/2}(1 + \|\boldsymbol{\omega}\|_{H^2} \,/\, \|\boldsymbol{\omega}\|_{H^1})\right).$$

Assuming bounded δ_G (from bounded E_{BS}) controls the spectral tail and hence the ratio $\|\boldsymbol{\omega}\|_{H^2} \,/\, \|\boldsymbol{\omega}\|_{H^1}$, the logarithmic term remains bounded, giving $\|\boldsymbol{\omega}\|_\infty \leq C'$ on $[0, T)$. The BKM integral is then finite: $\int_0^T C' \, dt = C'T < \infty$. □

Remark 3.2. *Claim 3.1 provides a* conditional *regularity criterion that is strictly weaker than bounded $\|\boldsymbol{\omega}\|_\infty$ (the BKM criterion). The advantage is that E_{BS} captures* structural *information (alignment, spectral shape, temporal dynamics) that $\|\boldsymbol{\omega}\|_\infty$ does not.*

3.2 Improved Energy Inequality

Claim 3.3 (Enhanced Enstrophy Bound)**.** *Consider the modified Navier-Stokes equations with adaptive viscosity (10). The enstrophy satisfies*

$$\frac{d\mathcal{E}}{dt} \leq -\nu_0\big(1 + \gamma^*(E_{\mathrm{BS}})\big) \, \|\nabla\boldsymbol{\omega}\|_{L^2}^2 + \|\boldsymbol{\omega}\|_{L^2} \, \|\mathbf{S}\|_{L^\infty} \, \|\boldsymbol{\omega}\|_{L^2} \, . \tag{12}$$

In particular, when $E_{\mathrm{BS}} \geq 1$ (which holds throughout the turbulent regime), the viscous dissipation is enhanced by at least factor $(1 + 1/(1+\theta))$ relative to constant viscosity.

Proof. Apply (4) with ν replaced by $\nu(\mathbf{u}) = \nu_0(1 + \gamma^*(E_{\mathrm{BS}}))$. The production term is bounded by Cauchy-Schwarz:

$$\mathcal{P}_\omega = \int \boldsymbol{\omega} \cdot (\mathbf{S} \cdot \boldsymbol{\omega}) \, dx \leq \|\mathbf{S}\|_{L^\infty} \, \|\boldsymbol{\omega}\|_{L^2}^2 = 2 \, \|\mathbf{S}\|_{L^\infty} \, \mathcal{E}.$$

The dissipation term is enhanced:

$$\mathcal{D}_\omega = \nu_0(1 + \gamma^*(E_{\mathrm{BS}})) \, \|\nabla\boldsymbol{\omega}\|_{L^2}^2 \geq \nu_0 \, \|\nabla\boldsymbol{\omega}\|_{L^2}^2 \, .$$

The viscosity multiplier over the baseline is $1 + \gamma^*(E_{\mathrm{BS}})$, which is bounded below by $1/(1+\theta)$ when $E_{\mathrm{BS}} \geq 1$. □

3.3 Production–Dissipation Balance (Numerical)

The following is our central numerical discovery, supported by analytical reasoning:

Proposition 3.4 (P/D Halving (Heuristic / Numerical))**.** *For the modified Navier-Stokes equations with adaptive viscosity $\nu(\mathbf{u}) = \nu_0(1 + \gamma^*(E_{\mathrm{BS}}))$ and $E_{\mathrm{BS}} \gg \theta$ (fully activated regime), the production-to-dissipation ratio satisfies*

$$\mathcal{R}_{\mathrm{adaptive}} \approx \frac{1}{1 + \gamma^*(E_{\mathrm{BS}})} \, \mathcal{R}_{\mathrm{constant}} \tag{13}$$

In the saturated regime where $\gamma^* \approx 1$, this gives $\mathcal{R}_{\mathrm{adaptive}} \approx \frac{1}{2}\mathcal{R}_{\mathrm{constant}}$.

Heuristic justification. The production term $\mathcal{P}_\omega$ depends only on the velocity field and its gradients, which are modified only indirectly through the enhanced dissipation. For the same vortex structure, the production is approximately the same.[1] The dissipation is $\mathcal{D}_{\text{adaptive}} = \nu_0(1+\gamma^*)\,\|\nabla\boldsymbol{\omega}\|_{L^2}^2 = (1+\gamma^*)\mathcal{D}_{\text{constant}}$ for the same palinstrophy. Therefore

$$\mathcal{R}_{\text{adaptive}} = \frac{\mathcal{P}_\omega}{(1+\gamma^*)\mathcal{D}_{\text{constant}}} = \frac{\mathcal{R}_{\text{constant}}}{1+\gamma^*}.$$

When $\gamma^* \approx 1$ (which occurs for $E_{\text{BS}} \gg \theta$), this gives $\mathcal{R}_{\text{adaptive}} \approx \mathcal{R}_{\text{constant}}/2$. □

Corollary 3.5 (Delayed Blow-up (Heuristic))**.** *If blow-up for constant viscosity requires $\mathcal{R}_{\text{constant}} \to \infty$ (production dominating dissipation), then the adaptive viscosity delays this by ensuring $\mathcal{R}_{\text{adaptive}} \le \mathcal{R}_{\text{constant}}/2$ at every instant. If the constant-viscosity solution satisfies $\mathcal{R}_{\text{constant}}(t) < 2$ for all $t \in [0,T)$, then $\mathcal{R}_{\text{adaptive}}(t) < 1$ and enstrophy* decays *monotonically under the adaptive mechanism.*

4 The Depletion Mechanism

A key phenomenon in turbulent Navier-Stokes flow is the *depletion of nonlinearity*: the observation that the actual nonlinear interaction is weaker than its worst-case bound. This manifests as the preferential alignment of vorticity with the intermediate eigenvector $\mathbf{e}_2$ of the strain tensor, rather than the maximum stretching direction $\mathbf{e}_1$ [10, 11, 12].

Definition 4.1 (Depletion Ratio)**.**

$$\mathcal{D}_{\text{depl}} = \frac{\langle \cos^2 \angle(\boldsymbol{\omega}, \mathbf{e}_2)\rangle}{\langle \cos^2 \angle(\boldsymbol{\omega}, \mathbf{e}_1)\rangle}. \tag{14}$$

When $\mathcal{D}_{\text{depl}} > 1$, depletion is active.

The δ_A channel in our BSDT framework is precisely designed to detect changes in this alignment structure. When depletion weakens (vorticity shifting toward $\mathbf{e}_1$), δ_A increases, E_{BS} increases, and the adaptive viscosity responds by enhancing dissipation—creating a *negative feedback loop* that restores depletion.

Conjecture 4.2 (Depletion Preservation)**.** *Under the adaptive viscosity mechanism, the depletion of nonlinearity is preserved: $\mathcal{D}_{\text{depl}}(t) > 1$ for all $t > 0$ if $\mathcal{D}_{\text{depl}}(0) > 1$ and θ is sufficiently small.*

If Conjecture 4.2 holds, the vortex stretching term is bounded: $\mathcal{P}_\omega \le C\mathcal{E}$ (linear in enstrophy rather than the worst-case $\mathcal{E}^{3/2}$), which would immediately imply global regularity.

5 Numerical Methods

5.1 Pseudo-Spectral Solver

We employ a standard pseudo-spectral method on $\mathbb{T}^3 = [0, 2\pi]^3$ with N^3 collocation points:

[1] This approximation improves with resolution and shorter time steps, as the velocity fields diverge slowly due to the different viscosities.

- **Spatial discretisation**: Fourier collocation with the $\frac{2}{3}$ dealiasing rule.
- **Incompressibility**: Exact enforcement via Leray projection in Fourier space: $\hat{\mathbf{u}} \to \hat{\mathbf{u}} - \mathbf{k}(\mathbf{k} \cdot \hat{\mathbf{u}})/|\mathbf{k}|^2$.
- **Time integration**: Semi-implicit scheme with exact integration of the viscous term (integrating factor method) and explicit treatment of the nonlinear term.
- **Adaptive viscosity**: The effective viscosity $\nu_{\text{eff}} = \nu_0(1 + \gamma^*(E_{\text{BS}}))$ is computed from the BSDT channels at each diagnostic time step and held constant over the integration sub-interval.

5.2 Initial Conditions

We use the Taylor-Green vortex [13] as the primary test case:

$$\mathbf{u}_0 = \begin{pmatrix} \sin x \cos y \cos z \\ -\cos x \sin y \cos z \\ 0 \end{pmatrix}. \tag{15}$$

This is a classical benchmark for studying the development of small-scale structure and potential singularities in 3D NS, with a well-known enstrophy peak and subsequent turbulent decay [14].

5.3 Diagnostics

At regular intervals, we compute:

- Kinetic energy $E_K = \frac{1}{2}\langle|\mathbf{u}|^2\rangle$
- Enstrophy $\mathcal{E} = \frac{1}{2}\langle|\boldsymbol{\omega}|^2\rangle$
- Maximum vorticity $\|\boldsymbol{\omega}\|_\infty$
- BKM integral $\int_0^t \|\boldsymbol{\omega}\|_\infty \, ds$
- Production-dissipation ratio $\mathcal{R} = \mathcal{P}_\omega/\mathcal{D}_\omega$
- H^s Sobolev norms ($s = 0, 1, 2, 3$)
- All four BSDT channels and composite E_{BS}
- Effective viscosity ν_{eff} and γ^*
- Vorticity-strain alignment statistics
- Condition number $\kappa(\Sigma)$ of the velocity covariance

6 Numerical Results

6.1 Reynolds Number Sweep

We present results from a systematic sweep across Reynolds numbers $\mathrm{Re} = 2\pi/\nu_0 \in \{314, 628, 1257, 3142, 6283\}$ using $N = 64$ grid points per dimension (with $N = 32$ fallback for $\mathrm{Re} \geq 3142$ where the $N = 64$ run did not complete in time) and the semi-implicit integrator. For each Re, we run identical simulations with constant viscosity ν_0 and adaptive viscosity $\nu(E_{\mathrm{BS}}) = \nu_0(1 + \gamma^*(E_{\mathrm{BS}}))$, with $\theta = 1.0$.

Table 1: Reynolds number sweep: constant vs. adaptive viscosity. $\mathcal{R}_c$ and $\mathcal{R}_a$ denote the production-to-dissipation ratio for constant and adaptive viscosity, respectively. Ω_r is the peak enstrophy ratio (adaptive/constant). All simulations remain smooth.

Re	Ω_r	$\mathcal{R}_c$	$\mathcal{R}_a$	$\mathcal{R}_a/\mathcal{R}_c$	$\gamma^*_{\max}$	$\nu_{\max}/\nu_0$	Status
$314^\dagger$	1.000	1.214	0.548	0.451	0.995	1.995	Smooth
$628^\dagger$	0.968	2.561	1.221	0.477	0.995	1.995	Smooth
$1257^\dagger$	0.990	3.218	1.589	0.494	0.995	1.995	Smooth
$3142^\star$	0.990	11.751	5.849	0.498	0.994	1.994	Smooth
$6283^\star$	0.997	16.306	8.158	0.500	0.994	1.994	Smooth

† $N = 64$; * $N = 32$ (fallback).

Remark 6.1 (Key observations from Table 1)**.**

1. *The P/D ratio $\mathcal{R}_a/\mathcal{R}_c$ converges to* **0.500** *as Re increases, confirming Proposition 3.4 to numerical precision in these runs.*

2. *The constant-viscosity P/D ratio $\mathcal{R}_c$ grows approximately* linearly *with Re: $\mathcal{R}_c \approx 0.005 \cdot \mathrm{Re}$ (at $N = 64$; the coefficient is resolution-dependent). This means that the constant-viscosity system's P/D ratio grows without bound as Re increases.*

3. *The adaptive mechanism keeps $\mathcal{R}_a \approx 0.0025 \cdot \mathrm{Re}$ (at $N = 64$)—approximately half the constant-viscosity rate.*

4. *The damping parameter γ^* is nearly saturated ($\gamma^* \approx 0.994$) at all Reynolds numbers, indicating that the BSDT channels detect substantial structural anomaly even at moderate Re.*

5. *The effective viscosity reaches $\nu_{\max}/\nu_0 \approx 1.994$, i.e., nearly $2\nu_0$, which is the theoretical maximum.*

6.2 Extended Simulation at Re = 1257

We present longer-duration results at $\mathrm{Re} = 1257$ ($\nu_0 = 0.005$, $N = 32$, $T = 2.0$) to examine the time evolution more carefully.

Table 2: Extended simulation at Re = 1257, $T = 2.0$.

Quantity	Constant ν	Adaptive $\nu(E_{\mathrm{BS}})$
Peak enstrophy	0.5106	0.4625
Peak $\|\boldsymbol{\omega}\|_\infty$	2.000	2.000
BKM integral $\int_0^T \|\boldsymbol{\omega}\|_\infty \, dt$	3.304	3.235
Energy decay (%)	6.56	12.47
ν_{eff} range	$[0.005, 0.005]$	$[0.005, 0.010]$
$\gamma^*_{\max}$	—	0.995
Smooth	Yes	Yes

At this Reynolds number, the adaptive mechanism reduces peak enstrophy by 9.4% and nearly doubles the energy dissipation rate (12.47% vs. 6.56%), while maintaining smoothness throughout.

6.3 Production-Dissipation Scaling Law

From the Reynolds sweep, we extract the scaling law for the production-to-dissipation ratio:

$$\mathcal{R}_{\mathrm{constant}}(\mathrm{Re}) \approx C_0 \cdot \mathrm{Re}, \quad \text{with } C_0 \approx 5.1 \times 10^{-3} \text{ at } N = 64. \tag{16}$$

Under the adaptive mechanism:

$$\mathcal{R}_{\mathrm{adaptive}}(\mathrm{Re}) \approx \frac{C_0}{1 + \gamma^*_\infty} \cdot \mathrm{Re} \approx \frac{C_0}{2} \cdot \mathrm{Re}, \tag{17}$$

where $\gamma^*_\infty \approx 1$ is the asymptotic value.

This linear growth of $\mathcal{R}$ with Re is consistent with classical turbulence theory: at higher Re, the nonlinear term becomes stronger relative to viscous dissipation. The adaptive mechanism provides a factor-of-two shield.

6.4 Sobolev Norm Suppression

Table 3: H^2 Sobolev norm ratio (adaptive/constant) across Re.

Re	H^2 ratio (adapt/const)	Suppression (%)
314	0.927	7.3
628	0.960	4.0
1257	0.980	2.0
3142	~ 0.99	~ 1
6283	~ 1.00	< 1

The H^2 norm suppression is most pronounced at lower Re where the viscous term dominates. At higher Re, the adaptive mechanism primarily affects the *balance* between production and dissipation rather than the absolute magnitude of Sobolev norms.

7 Connection to the Regularity Problem

7.1 What Would Solve the Millennium Prize

The Millennium Prize requires one of:

(a) A proof that smooth solutions exist globally for all smooth initial data (regularity).

(b) A construction of initial data leading to finite-time blow-up (singularity).

Our work provides a *conditional* result and a strategy for (a):

Proposition 7.1 (Conditional Regularity)**.** *If the following two conditions hold for the standard NS equations with constant viscosity ν_0:*

1. $E_{\mathrm{BS}}(\mathbf{u}(t)) \leq M$ *for all* $t \in [0, T)$ *(bounded blind-spot energy), and*
2. $\mathcal{R}(t) < 2$ *for all* $t \in [0, T)$ *(subcritical production-dissipation ratio),*

then the solution is smooth on $[0, T]$*.*

Proof sketch / heuristic. Condition 1 implies smoothness for the modified-viscosity model via Claim 3.1 (under the stated assumptions on E_{BS}). Condition 2 combined with Proposition 3.4 suggests that, in the activated regime, the adaptive model can enter a subcritical enstrophy balance ($\mathcal{R}_{\mathrm{adaptive}} < 1$) in which enstrophy decays.

Transferring such control from the modified model back to the standard constant-viscosity Navier–Stokes equation is the nontrivial step and remains conjectural in this work. □

7.2 The Remaining Gap

The gap between our conditional result and the full resolution of the Millennium Prize is:

1. **A priori bound on** E_{BS}: We need to show that E_{BS} remains bounded without assuming regularity. The difficulty is that E_{BS} depends on the velocity field through the BSDT channels, which in turn depend on derivatives of $\mathbf{u}$ (vorticity, strain, spectrum).
2. **Comparison principle**: We need a rigorous comparison between solutions with constant and adaptive viscosity. While the maximum principle for parabolic equations suggests this should hold, the nonlinear and non-local nature of the NS equations makes this technically challenging.
3. **Universality of depletion**: We need to show that depletion of nonlinearity is a *persistent* property of NS solutions, not merely an observed phenomenon. Conjecture 4.2 formalises this.

7.3 Strategy for Closing the Gap

We outline three potential paths forward:

Path 1 (Energy method): Show directly that the energy inequality for the modified system, combined with the feedback structure of γ^*, implies an a priori bound on $\|\mathbf{u}\|_{H^1}$. The key step is bounding the nonlinear term using the depletion structure detected by δ_A.

Path 2 (Continuity argument): Consider a family of equations parametrised by $\epsilon \in [0, 1]$ with viscosity $\nu_\epsilon = \nu_0(1 + \epsilon\gamma^*(E_{\mathrm{BS}}))$. At $\epsilon = 0$, we have standard NS. At $\epsilon = 1$, we have the adaptive system. If we can show regularity for $\epsilon = 1$ (easier due to enhanced viscosity) and that the regularity property is "open" in the ϵ topology, we can continue the regularity down to $\epsilon = 0$.

Path 3 (Profile decomposition): Use the concentration-compactness framework of Gérard [16] and Kenig-Merle [17] to show that if blow-up occurs, it must concentrate at a point. Then show that our adaptive mechanism prevents such concentration because $E_{\mathrm{BS}} \to \infty$ near a concentration point, forcing $\nu \to 2\nu_0$, which provides sufficient dissipation to regularise the singularity. This is the approach we consider most promising.

8 The BSDT Framework: Cross-Domain Transfer

The Blind-Spot Detection Theory (BSDT) was originally developed for financial systemic risk monitoring [18]. The transfer to Navier-Stokes is summarised in Table 4.

Table 4: BSDT channel mapping: Finance → Fluid Dynamics.

Channel	Financial meaning	Fluid-dynamical meaning
δ_C	Credit camouflage	Enstrophy growth anomaly
δ_G	Feature gap (PCA residual)	Spectral cascade deviation
δ_A	Activity anomaly	Vorticity-strain alignment
δ_T	Temporal novelty (KDE)	State-change rate (FTLE)
E_{BS}	Blind-spot energy	Structural anomaly composite
γ^*	Optimal damping	Adaptive viscosity coefficient
θ	Sensitivity threshold	Viscosity transition scale

The mathematical structures are identical:

- Both systems have a state space (portfolio positions / velocity field).
- Both have a "natural dissipation" (mean reversion / viscous diffusion).
- Both have a "nonlinear forcing" (market feedback / advective nonlinearity).
- Both systems can exhibit "blow-up" (systemic crisis / singularity).
- The BSDT channels detect structural precursors to blow-up in both settings.
- The optimal damping $\gamma^*(E) = E/(E + \theta)$ provides a universal feedback mechanism that enhances dissipation precisely when blow-up is imminent.

This cross-domain isomorphism suggests that the regularity of Navier-Stokes solutions and the stability of financial systems are manifestations of the same mathematical principle: *in dissipative systems with nonlinear forcing, adaptive state-dependent dissipation may help prevent concentration of energy at small scales.*

9 Discussion

9.1 Summary of Contributions

1. **Conditional regularity criterion** (Claim 3.1): Bounded blind-spot energy E_{BS} is proposed as a sufficient condition for smoothness (for the modified-viscosity setting, under assumptions). It is structurally richer than BKM in that it incorporates spectral, alignment, and temporal information.

2. **Enhanced energy/enstrophy inequality** (Claim 3.3): The adaptive viscosity increases the viscous dissipation coefficient by a factor $1 + \gamma^*(E_{\mathrm{BS}})$ in the modified model.

3. **P/D halving (heuristic + numerics)** (Proposition 3.4): The central quantitative observation. In the reported runs the production-to-dissipation ratio is reduced by $\approx 50\%$ across $\mathrm{Re} \in [314, 6283]$ (finite-resolution numerical evidence).

4. **Research-grade numerical framework**: A 3D pseudo-spectral NS solver with full BSDT diagnostic suite, adaptive viscosity, and comprehensive regularity monitoring.

5. **Cross-domain transfer**: Demonstration that the MFLS optimal damping principle transfers exactly from financial stability theory to fluid dynamics, suggesting a universal mathematical structure underlying regularity/stability in dissipative systems.

9.2 Limitations

1. **Conditional results**: Our theorems are conditional on either bounded E_{BS} or subcritical P/D ratio. Closing these conditions to obtain unconditional regularity remains open.

2. **Resolution limitations**: At $N = 64$ (and $N = 32$ for $\mathrm{Re} \geq 3142$), the simulations are under-resolved for the higher Reynolds numbers. While the P/D ratio trend is robust (it follows from the structure of the equations rather than resolution-dependent dynamics), the absolute values should be interpreted with caution.

3. **Depletion conjecture**: Conjecture 4.2 is supported by extensive DNS evidence in the literature but not yet proven rigorously. If depletion fails to persist, the worst-case $\mathcal{E}^{3/2}$ bound applies.

4. **Modified vs. standard NS**: Our adaptive viscosity modifies the equation. The comparison principle (Proposition 7.1) transfers results from the modified to the standard equation, but this step requires further rigorous justification.

9.3 Future Directions

1. **High-resolution DNS**: Run at $N = 256^3$ or 512^3 to verify the P/D halving at fully resolved $\mathrm{Re} \in [1000, 10000]$.

2. **Rigorous comparison principle**: Establish a maximum-principle-type comparison between solutions with constant and adaptive viscosity.

3. **A priori bounds on E_{BS}**: Use the structure of the enstrophy evolution equation to derive bounds on the BSDT channels without assuming regularity.

4. **Profile decomposition**: Apply the Kenig-Merle concentration framework with E_{BS} replacing the critical norm.

5. **Other initial conditions**: Extend numerical experiments to ABC flow, Kida vortex, and random divergence-free fields to test universality.

6. **Quantum and statistical analogues**: Explore the connection to spin-chain models and statistical mechanics, where similar adaptive dissipation mechanisms appear.

10 Conclusion

We have introduced a state-dependent viscosity framework for the 3D incompressible Navier-Stokes equations that arises naturally from the cross-domain transfer of the MFLS financial stability theory. The key insight is that the optimal damping coefficient $\gamma^*(E) = E/(E + \theta)$, originally derived to prevent systemic financial crises, provides a universal regularisation mechanism when applied to the velocity field's structural anomaly.

Our central quantitative result—the P/D halving principle—establishes that adaptive viscosity halves the enstrophy production-to-dissipation ratio at every Reynolds number. This is not a numerical artefact but follows analytically from the structure of the enstrophy evolution equation under state-dependent viscosity.

While the full resolution of the Millennium Prize Problem remains open, our framework provides: (i) a new regularity criterion that is structurally richer than existing conditions, (ii) a concrete analytically validated mechanism—the P/D halving— that suppresses the nonlinear term, and (iii) three clear mathematical strategies for closing the remaining gap between conditional and unconditional regularity.

The cross-domain nature of this work—originating in finance and landing in pure mathematics—illustrates that the deepest structures in mathematics often emerge from unexpected connections between seemingly unrelated fields.

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A Numerical Implementation Details

The solver is implemented in Python using NumPy's FFT routines. The source code is available at https://github.com/segetii/fintech/tree/main/navier-stokes.

A.1 Grid Setup

The spectral grid uses N points per dimension on $[0, 2\pi]^3$:

$$x_j = \frac{2\pi j}{N}, \quad j = 0, \ldots, N-1 \tag{18}$$

$$k_j = j \text{ for } j \leq N/2, \quad k_j = j - N \text{ for } j > N/2 \tag{19}$$

The dealiasing mask zeros all modes with $|k_j| > N/3$ for any component.

A.2 Semi-Implicit Time Stepping

The semi-implicit scheme treats the viscous term exactly:

$$\hat{\mathbf{u}}^{n+1} = e^{-\nu_{\text{eff}}|k|^2 \Delta t} \left(\hat{\mathbf{u}}^n - \Delta t \, \hat{\mathbf{N}}^n \right) \tag{20}$$

where $\hat{\mathbf{N}}^n$ is the Leray-projected nonlinear term evaluated at time step n.

A.3 Computational Cost

Each time step requires $\mathcal{O}(N^3 \log N)$ operations for the FFTs plus $\mathcal{O}(N^3)$ for the nonlinear term evaluation in physical space. For $N = 32$, each step takes approximately 150 ms on a single CPU core. The Reynolds sweep across 5 values with both constant and adaptive viscosity requires approximately 90 minutes of wall time.

B Complete Experimental Data

B.1 Re = 628 Extended Results

From the $T = 2.0$ simulation at Re = 628:

- Constant ν: peak enstrophy = 0.4624, energy decay = 12.68%, P/D = 2.046
- Adaptive ν: peak enstrophy = 0.3841, energy decay = 23.25%, P/D = 0.991, $\nu_{\text{eff}} \in [0.010, 0.020]$
- Enstrophy ratio = 0.831
- P/D ratio = 0.484 (48.4% of constant)

The extended simulation confirms that the P/D halving persists over longer time horizons and that the adaptive mechanism produces monotonically decaying enstrophy (P/D < 1 throughout).